CS 215: Task 06: November 11, 2026

1. The time taken by non-hostel resident students to travel to IIT Bombay has mean 42 minutes and standard deviation 12 minutes. Travel times on different days may be treated as independent.

   Suppose the travel times over 64 days are recorded. What is the probability that the average travel time is at least 39 minutes and at most 45 minutes?

   Solution: No assumption is made about the distribution of the travel time on an individual day.

   $$\overline{X} = \frac{1}{64}\sum_{i=1}^{64} X_i$$

   denotes the average travel time. Assumption: Travel time on the 64 days are iid with mean and variance as stated $(42, 12^2)$.

   Using the Central Limit Theorem, we approximate

   $$P(39 \le \overline{X} \le 45).$$

   We have

   $$E[X_i] = 42, \qquad \mathrm{Var}(X_i) = 12^2.$$

   Therefore,

   $$E[\overline{X}] = 42, \qquad \mathrm{Var}(\overline{X}) = \frac{12^2}{64}.$$

   For $n = 64$, the Central Limit Theorem gives

   $$\overline{X} \approx \mathcal{N}\left(42, \frac{12^2}{64}\right).$$

   Thus,

   $$\mathrm{sd}(\overline{X}) = \frac{12}{8} = 1.5.$$

   Hence,

   $$\begin{aligned} P(39 \le \overline{X} \le 45) &\approx P\left(\frac{39-42}{1.5} \le Z \le \frac{45-42}{1.5}\right) \\ &= P(-2 \le Z \le 2). \end{aligned}$$

   Using the standard normal table,

   $$P(-2 \le Z \le 2) = 2\Phi(2) - 1 \approx 2(0.9772) - 1.$$

   Therefore,

   $$\boxed{P(39 \le \overline{X} \le 45) \approx 0.9544.}$$

2. The amount spent by a typical customer at Cafe92 has mean Rs. 180 and a standard deviation Rs. 90. Amounts spent by different customers may be treated as independent. On a particular evening, 100

customers visit Cafe92. What is the probability in percentage that the total revenue exceeds Rs. 20,000?

Solution: Let

$$T = \sum_{i=1}^{100} X_i$$

denote the total amount spent by these customers where

$$X_1, \ldots, X_{100} \text{ are assumed independent and identically distributed,}$$

with

$$E[X_i] = 180, \qquad \text{Var}(X_i) = 90^2.$$

Using the Central Limit Theorem, we approximate

$$P(T > 20000).$$

For each customer,

$$E[X_i] = 180, \qquad \text{Var}(X_i) = 90^2.$$

Therefore,

$$E[T] = 100(180) = 18000,$$

and, using independence,

$$\text{Var}(T) = 100(90^2).$$

Thus,

$$\text{sd}(T) = 90\sqrt{100} = 900.$$

By the Central Limit Theorem,

$$T \approx \mathcal{N}(18000, 900^2).$$

Hence,

$$P(T > 20000) \approx P\left(Z > \frac{20000 - 18000}{900}\right)$$
$$= P(Z > 2.22).$$

Using the standard normal table,

$$P(Z > 2.22) = 1 - \Phi(2.22) \approx 1 - 0.9868.$$

Therefore,

$$\boxed{P(T > 20000) \approx 0.0132 \text{ i.e. } 1.32\%.}$$

3. For each student in a class, independently, the probability of attending an optional review session is 0.62. A total of 400 students are eligible to attend. Using the Central Limit Theorem, approximate the probability that the number of students attending is between 240 and 260, inclusive.

Solution:

Let W denote the number of students who attend the session.

Using a normal approximation with continuity correction, we want to approximate

$$P(240 \le W \le 260).$$

where

$$W \sim \text{Binomial}(400, 0.62).$$

Therefore,

$$E[W] = 400(0.62) = 248,$$

and

$$\text{Var}(W) = 400(0.62)(0.38) = 94.24.$$

Thus,

$$\text{sd}(W) = \sqrt{94.24} \approx 9.71.$$

Using continuity correction,

$$P(240 \le W \le 260) \approx P(239.5 < W < 260.5).$$

Therefore,

$$\begin{aligned} P(240 \le W \le 260) &\approx P\left(\frac{239.5 - 248}{9.71} < Z < \frac{260.5 - 248}{9.71}\right) \\ &\approx P(-0.88 < Z < 1.29). \end{aligned}$$

Using the standard normal table,

$$\begin{aligned} P(-0.88 < Z < 1.29) &= \Phi(1.29) - \Phi(-0.88) \\ &= 0.9015 - (1 - 0.8106) \\ &= 0.7121. \end{aligned}$$

Hence,

$$\boxed{P(240 \le W \le 260) \approx 0.712.}$$

4. The lifetime of an electronic chip has mean 500 hours and standard deviation 20 hours. Lifetimes of different chips may be treated as independent. The distribution of the lifetime of an individual component is not assumed to be normal.

   A student wishes to purchase n independently selected chips. Let $\overline{X}$ denote their average lifetime. Using the Central Limit Theorem, determine the smallest integer n for which

$$P\left(|\overline{X} - 500| < 5\right) \ge 0.95.$$

Teaching-team notes.

$$P(|Z| \le 1.96) \approx 0.95, \qquad Z \sim \mathcal{N}(0, 1).$$

You are expected to remember this number 1.96 during the exam, much the way you remember $\pi \approx 3.14$. It corresponds to the 95% confidence interval.

Solution: For a sample of size n,

$$E[\overline{X}] = 500$$

and

$$\mathrm{sd}(\overline{X}) = \frac{20}{\sqrt{n}}.$$

By the Central Limit Theorem,

$$\overline{X} \approx \mathcal{N}\left(500, \frac{20^2}{n}\right).$$

We require

$$P\left(|\overline{X} - 500| < 5\right) \geq 0.95.$$

Using

$$P(|Z| \leq 1.96) \approx 0.95,$$

we require

$$\frac{5}{20/\sqrt{n}} \geq 1.96.$$

Therefore,

$$\frac{\sqrt{n}}{4} \geq 1.96,$$

so

$$\sqrt{n} \geq 7.84.$$

Squaring,

$$n \geq 61.4656.$$

Since n must be an integer, the smallest possible value (using the CLT) is

$$\boxed{n = 62}.$$

5. The weight of a packaged item has mean 100 grams and standard deviation 24 grams. Weights of different items may be treated as independent. A distributor is allowed to return a shipment back to the manufacturer if the average weight of the items in the shipment is more than c since the consumer gets more quantity for the same price.

   A distributor handles a random shipment of 144 items. Let $\overline{X}$ denote the average weight of the items. Using the Central Limit Theorem, find the value c such that approximately

   $$P(\overline{X} \leq c) = 0.975.$$

Teaching-team notes. You may use

$$P(Z \leq 1.96) = 0.975, \qquad Z \sim \mathcal{N}(0, 1).$$

Solution: We have

$$E[\overline{X}] = 100$$

and

$$\mathrm{sd}(\overline{X}) = \frac{24}{\sqrt{144}} = 2.$$

By the Central Limit Theorem,

$$\overline{X} \approx \mathcal{N}(100, 2^2).$$

We want

$$P(\overline{X} \leq c) = 0.975.$$

Since

$$P(Z \leq 1.96) = 0.975,$$

we require

$$\frac{c - 100}{2} = 1.96.$$

Thus,

$$c - 100 = 3.92,$$

and hence

$$\boxed{c = 103.92 \text{ grams.}}$$

6. Suppose that, independently for each student, the probability that a student uses public transport to commute to campus is 0.40.

   A random sample of 600 students is chosen. Let $\widehat{P}$ denote the proportion of students in the sample who use public transport.

   Using the Central Limit Theorem for proportions, approximate

$$P(0.36 \leq \widehat{P} \leq 0.44).$$

Solution: Let X_i be an indicator variable that denotes 1 if student i uses public transport. Then

$$\widehat{P} = \sum_{1}^{600} X_i/600$$

We have

$$E[\widehat{P}] = 0.40$$

and

$$\text{Var}(\widehat{P}) = \frac{(0.40)(0.60)}{600} = 0.0004.$$

Therefore,

$$\text{sd}(\widehat{P}) = 0.02.$$

Using the Central Limit Theorem,

$$\widehat{P} \approx \mathcal{N}(0.40, 0.02^2).$$

Hence,

$$P(0.36 \leq \widehat{P} \leq 0.44) \approx P\left(\frac{0.36 - 0.40}{0.02} \leq Z \leq \frac{0.44 - 0.40}{0.02}\right)$$
$$= P(-2 \leq Z \leq 2).$$

Using the standard normal table,

$$P(-2 \leq Z \leq 2) = 2\Phi(2) - 1 = 2(0.9772) - 1.$$

Thus,

$$\boxed{P(0.36 \leq \widehat{P} \leq 0.44) \approx 0.9544.}$$

7. For each of the following cases, consider

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

For each case, determine whether the Central Limit Theorem as stated in class can be applied to approximate the distribution of $\overline{X}_n$ for large n. If it can, give the approximate distribution. If it cannot, state which assumption is missing. Write your answer starting with a phrase such as "Applies".

(a) The random variables $X_1, \ldots, X_n$ are i.i.d. with

$$E[X_i] = 5, \qquad \mathrm{Var}(X_i) = 9.$$

Solution:

Applies

The random variables are independent and identically distributed, and have finite mean and finite variance. Therefore, the Central Limit Theorem applies.

For large n,

$$\boxed{\overline{X}_n \approx \mathcal{N}\left(5, \frac{9}{n}\right).}$$

(b) The random variables $X_1, \ldots, X_n$ are i.i.d. with

$$E[X_i] = 5, \qquad E[X_i^2] = \infty.$$

Solution:

Does not apply.

Although the mean exists, the second moment is infinite, so the finite variance condition required by the version of the Central Limit Theorem used in class is not satisfied.

Hence, this version of the Central Limit Theorem cannot be applied.

(c) The random variables $X_1, \ldots, X_n$ are independent and satisfy

$$E[X_i] = 5, \qquad \mathrm{Var}(X_i) = 9,$$

but they are not assumed to have the same distribution.

Solution: Cannot conclude authoritatively.

The random variables are independent and have the same mean and variance, but they are not assumed to be identically distributed.

Therefore, the i.i.d. version of the Central Limit Theorem stated in class cannot be directly applied.

Does not necessarily apply. Additional conditions that may allow non-identical random variables has not been stated in the question.

(d) The random variables $X_1, \ldots, X_n$ are i.i.d. with

$$X_i \sim \mathcal{N}(5, 9).$$

In this case, also state whether the distribution obtained for $\overline{X}_n$ is approximate or exact.

Solution:

Applies, not necessary.

Since the X_i are independent normal random variables, their average is itself normally distributed. Thus,

$$\boxed{\overline{X}_n \sim \mathcal{N}\left(5, \frac{9}{n}\right).}$$

This result is exact for every n; it does not require a large-n approximation.

8. Let $X_1, X_2, \ldots$ be i.i.d. random variables satisfying

$$E[X_i] = \mu, \qquad \mathrm{Var}(X_i) = \sigma^2,$$

and suppose

$$E[(X_i - \mu)^3] = \gamma\sigma^3,$$

where $\gamma \neq 0$.

For

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i,$$

define the standardized sample mean

$$Z_n = \frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma}.$$

(a) Show that

$$E[Z_n] = 0, \qquad \mathrm{Var}(Z_n) = 1.$$

Solution: Let

$$Y_i = X_i - \mu.$$

Then

$$E[Y_i] = 0, \qquad E[Y_i^2] = \sigma^2, \qquad E[Y_i^3] = \gamma\sigma^3.$$

Also,

$$Z_n = \frac{1}{\sigma\sqrt{n}}\sum_{i=1}^{n} Y_i.$$

Using linearity of expectation,

$$E[Z_n] = \frac{1}{\sigma\sqrt{n}}\sum_{i=1}^{n} E[Y_i] = 0.$$

Using independence,

$$\begin{aligned}
\mathrm{Var}(Z_n) &= \frac{1}{\sigma^2 n}\,\mathrm{Var}\left(\sum_{i=1}^{n} Y_i\right) \\
&= \frac{1}{\sigma^2 n}\sum_{i=1}^{n}\mathrm{Var}(Y_i) \\
&= \frac{n\sigma^2}{n\sigma^2} \\
&= 1.
\end{aligned}$$

Hence,

$$\boxed{E[Z_n] = 0, \qquad \mathrm{Var}(Z_n) = 1.}$$

(b) Show that

$$E[Z_n^3] = \frac{\gamma}{\sqrt{n}}.$$

Solution: We have

$$E[Z_n^3] = \frac{1}{\sigma^3 n^{3/2}} E\left[\left(\sum_{i=1}^{n} Y_i\right)^3\right].$$

When the cube is expanded, all mixed terms have expectation zero because the Y_i are independent and have mean zero. Therefore,

$$E\left[\left(\sum_{i=1}^{n} Y_i\right)^3\right] = \sum_{i=1}^{n} E[Y_i^3].$$

Thus,

$$\begin{aligned} E[Z_n^3] &= \frac{n\gamma\sigma^3}{\sigma^3 n^{3/2}} \\ &= \boxed{\frac{\gamma}{\sqrt{n}}}. \end{aligned}$$

(c) Compute $E[Z_n^3]$ for

$$n = 10, \qquad n = 200, \qquad n = 1000$$

and compare their magnitudes.

Solution: For the three sample sizes,

$$E[Z_{10}^3] = \frac{\gamma}{\sqrt{10}},$$

$$E[Z_{200}^3] = \frac{\gamma}{\sqrt{200}},$$

and

$$E[Z_{1000}^3] = \frac{\gamma}{\sqrt{1000}}.$$

Therefore,

$$\boxed{\left|E[Z_{1000}^3]\right| < \left|E[Z_{200}^3]\right| < \left|E[Z_{10}^3]\right|.}$$

The third standardized moment of a normal random variable is 0. Thus, this measure of asymmetry decreases towards zero as the sample size increases.

(d) Now take five independent batches, each containing 200 iid observations from the same distribution as X_i. Let $\overline{X}_{200}^{(j)}$ be the sample mean of batch j, and define

$$M = \frac{1}{5}\sum_{j=1}^{5} \overline{X}_{200}^{(j)}.$$

Show that M has the same mean, variance, and standardized third moment as a sample mean based on 1000 observations.

Solution: Write the observations in batch j as

$$X_{1j}, X_{2j}, \ldots, X_{200j}.$$

Then

$$\overline{X}_{200}^{(j)} = \frac{1}{200}\sum_{i=1}^{200} X_{ij}.$$

Hence,

$$\begin{aligned} M &= \frac{1}{5}\sum_{j=1}^{5} \overline{X}_{200}^{(j)} \\ &= \frac{1}{5}\sum_{j=1}^{5}\left(\frac{1}{200}\sum_{i=1}^{200} X_{ij}\right) \\ &= \frac{1}{1000}\sum_{j=1}^{5}\sum_{i=1}^{200} X_{ij}. \end{aligned}$$

Thus, M is simply the sample mean of all 1000 observations.

Therefore,

$$E[M] = \mu,$$

and

$$\mathrm{Var}(M) = \frac{\sigma^2}{1000}.$$

If

$$Z_M = \frac{\sqrt{1000}(M-\mu)}{\sigma},$$

then, by the result of part (b),

$$E[Z_M^3] = \boxed{\frac{\gamma}{\sqrt{1000}}}.$$

Thus, averaging five independent sample means based on 200 observations each gives the same mean, variance, and standardized third moment as taking one sample mean based on all 1000 observations.